



Holding onto the Magic:

Lightweight Augmentation of Digital Reading Devices

Emma Thom and Matt Jones
 Future Interaction Technology Lab
 Swansea University
 csemma, matt.jones @swansea.ac.uk

ABSTRACT

Reading devices such as the Kindle are becoming widely used and many users are now routinely reading on tablet computers. The physical form factor of these devices promotes and accommodates comfortable, natural interactions, providing some of the look and feel of their paper predecessors. However, for operations such as searching and cutting and pasting, the magic and charm of paper is lost and the user has to revert to device-centred selections and menu operations. In this paper, we introduce and explore an alternative set of approaches. In the *NoteDrop* system we use hand-only gestures along with haptic and limited visual feedback. The aim is to enhance the active reading process with a focus on interactions with digital research documents. In addition to describing the concept prototype, we present the findings of an exploratory user-study that highlights the value and challenges of these new methods.

Author Keywords

Active reading, spatial interaction, pick up and place.

ACM Classification Keywords

H.5.2 [User Interfaces]: Input devices and strategies; Interaction styles; Haptic I/O.

General Terms

Design, Experimentation, Human Factors.

INTRODUCTION

Consider the commercial adverts for reading devices like the Kindle or tablets such as the iPad. Users are shown effortlessly perusing digital books, magazines and the like – relaxing in a comfy chair, often one-handed, just like the ways they are used to with the physical counterparts. Unlike paper documents, though, these devices can also provide digital features: users can, for instance, search or annotate their digital document; or cut and paste sections for later use. It is our view here, though, that in providing these sorts of services, current devices force the reader to shake

themselves from the magical and engaging experience to confront again a conventional, device-centred user interface: typing text, selecting from menus and so on.

In contrast, we are more interested in lighter weight augmentation. The *NoteDrop* concept introduced in this paper, couples a tablet reading device with a bracelet augmented with inertial sensors and actuators. The sensors are used to provide one-handed, fluid actions on the document: to cut and paste sections for later use; and to carry out search functions. Haptic actuators are used to give simple feedback both confirming the success of operations and alerting the reader to content of interest on the page. Additional lightweight feedback is provided by a coloured set of visual lights. This feedback helps the reader gain further insight into the text being read without having to break away from the document and – for instance – carry out a digital library search of research articles.

Consider the following scenario, which we explore:

Sam is lying on his couch at home, reading a research paper on his tablet computer. He sees a neat figure that he'd like to use and reference in one of his own articles. He grabs the image with his thumb and forefinger, picks it up and drops it to the left of his tablet, into the virtual clipboard that floats off-screen. The bracelet on his wrist buzzes gently to show the clipboard has received content.

A little later on, as he flicks through the article, the bracelet buzzes again – he pauses and reads the page a little more closely. “Ah”, he thinks, “lots of references to his supervisor’s work – must mention this to her”.

Returning to the article after a brief coffee break, Sam comes across an interesting idea referring to “tangible reading”. He wonders how prevalent this term is in the research community. Picking up the phrase with his fingers, he then drops it into the virtual search box in the space to the right of the tablet’s display. One of the four LED lights on the bracelet lights up – this indicates the term is found in documents indexed by Google Scholar.

Later that day, when back at his desk and on his PC, he reviews the papers retrieved by this and other searches he’s done that day along with the items on the clipboard.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee.

DIS 2012, June 11–15, 2012, Newcastle, UK.

Copyright 2012 ACM 978-1-4503-1210-3/12/06...\$10.00.

BACKGROUND

The active reading process involves making annotations in order to interpret the meaning of a document. Tablet-based prototypes exploring the digital support of this process include XLibris [9] and InkSeine [4]. Meanwhile, the Projective Augmented Book [1] explores projection onto a physical printed book in the style of a reading lamp. A digital pen captures gestures with specially printed paper and allows virtual annotations and projected traces. Using a pen gesture or via selecting a function from a palette, the user can perform various tasks on text such as dictionary look up, add to scrapbook, project annotations onto page and email. Any feedback is then projected onto the book.

The *NoteDrop* system explores interaction without marking documents and uses lightweight off screen feedback for search results. Unlike previous systems, *NoteDrop* aims to support the active reading process including marking, delayed reviewing [5] and in situ search, via minimal user interaction and without disturbing the natural reading process of the user.

Further to this, our design of “pick up and place” was inspired by Rekimoto [8]. Rekimoto explores several types of multiple finger interaction to control objects on interactive surfaces. This includes the pick up and dropping interaction. The notion of placing off screen was inspired by Gustafson et al. [3], with the aim being not to distract the user from the reading process. Other work has also considered the use of off-screen hand gestures with mobile phones and other surfaces, e.g., [6]. ReachMedia [2] explores seamlessly accessing information about physical objects, for example a book, via voice output and gesture input based on a wearable device.

Finally, our principles for the design of the interaction style include the term lightweight as described by Marshall and Bly [7]. The term considers subtle and invisible movements that a user experiences when turning a page in a physical document. This is often missing from digital devices.

NOTEDROP IMPLEMENTATION

To carry out an initial, exploratory study, *NoteDrop* has been implemented as follows. A Samsung Galaxy 7-inch tablet running Android 2.1 is used to display the document. In addition, at the top of the display, there are four icons (that can be lit or faded): these are used to represent the wearable LEDs mentioned in the scenario above. A wearable bracelet, based on a compact inertial sensor pack, captures spatial gestures and augments the content of the document with in situ and lightweight haptic highlighting. Haptic feedback was incorporated into the bracelet, allowing for future extension beyond digital documents. A Bluetooth connection was used for two-way communication between the bracelet and the tablet.

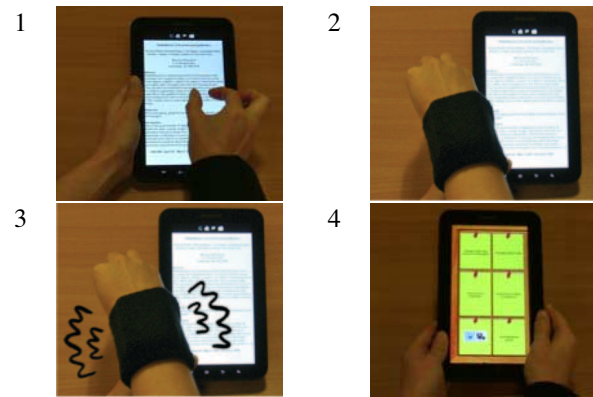


Figure 1: Adding notes to the clipboard. (1) Picking up an item; (2) Placing the item into the clipboard; (3) Vibrotactile feedback to confirm action; (4) Clipboard accessed on tablet showing previously gathered content.

Interaction Style

Using the concept of pick up and place *NoteDrop* allows users to “pick up” elements from a document and “place” them within the space surrounding the document. To implement the “pick up” feature, the multi-touch capabilities of the tablet are used to detect two points of contact with the screen. Elements between these two points are then taken to be the item of interest. Depending on the direction in which the element is placed (detected using the wearable bracelet), different functions of the system are initiated and various forms of lightweight feedback are given. Placing to the left of the document, i.e., into the “clipboard”, stores the selected elements for later use. Placing to the right of the document, i.e., into the “search box”, initiates a query of both external resources and elements within the document.

Clipboard

When an element is placed into the clipboard, it is stored and can be used for later reflection. To confirm that an element has been placed into the clipboard, a two-phased short vibration is initiated on the wearable device. The user can review notes added to the clipboard using the built-in review feature, accessible via an on-screen button. Interaction with the clipboard is illustrated in Figure 1.

Search Box

When an element is placed into the search box a query focused around the selected element is initiated. A short vibration is given on the wearable device to indicate successful placement of an element.

Two types of search were considered. The first approach explored giving *in situ* feedback in the form of a set of four small light-based indicators, which when lit, each show the outcome from a particular web resource (seen at the top of the display in Figure 1). Resources queried include: Google, Google Scholar and Google Images. For each search term, a query is made in turn to these services. The results of the

standard Google search are parsed and if the search term occurs in one of the top ten results, the “Google” light is lit. If there is at least one Wikipedia result containing the exact term then the “Wikipedia” light is lit. A similar process is carried out for the Scholar and Images services.

As an alternative to external search, *within* document search was also explored. In this case, when the search term occurs on the page currently in view, there is a vibration on the bracelet. While the current implementation explicitly links a search action to later bracelet feedback, we envisage automatic search and notification. For example by using knowledge of, say, concepts the reader is interested in to trigger feedback whenever these are mentioned on a page.

EXPLORING NOTEDROP

We wanted to assess the value of the prototype, gathering insights into alternative and additional interaction designs and to uncover challenges that impede the vision it embodies. To do this we recruited 22 active university researchers to take part in a lab study, aged from 22 to 56, from backgrounds outside of Computer Science and HCI. Fourteen participants were male and eight were female. Two of the male participants were left-handed. All participants except one had had prior use of a touch screen device. As we wanted to assess some variants on the design, we divided the 22 participants into two groups A and B, each containing an equal balance of left and right-handed users and genders. The two groups were used for comparisons in the **search task**, outlined below.

At the beginning of the study participants were introduced to *NoteDrop* and were given two minutes to familiarise themselves with the system. The system displayed a seven page published research document. The participants were then given a set of tasks to carry out, each of which involved a set of sub-tasks. Each task was followed by a discussion. At the end of the study, all participants were rewarded with a voucher for their time. Studies were video recorded and interactions with the system were logged.

For the **clipboard task** all participants from both groups were asked to pick up any three items (text or images) of interest from the document. They then placed them into the clipboard using the hand gesture. After this, they were shown the clipboard (see Figure 1).

Two types of **search task** were investigated. To explore the use of haptic highlighting, participants were asked to perform a search of eight pre-defined search terms within the document. Each word occurred between two and three times distributed evenly throughout the document. For Group A, the wearable device gave a short vibration when they turned to a page containing the search term (that is, exhibiting the haptic highlighting). For Group B, acting as a control, comparison group, the search term was highlighted

visually on the page in a similar manner to current PDF viewers. The time taken for a participant to find and touch occurrences of the terms was recorded. A discussion focused around the usefulness and potential use of the lightweight feedback then followed after the task.

To explore the **search-plus-light** feedback all participants from both groups were asked to pick-up three pre-defined search terms from a page in the document and to place them into the search box. After each interaction the participants were then asked to comment on the meaning and usefulness of the feedback provided. They were also asked what other information could be presented. Finally they were shown the results each light represented.

RESULTS

Pick Up and Place Technique

Throughout the various tasks we measured the accuracy of the system in detecting a pick up and then place action; and the accuracy of the elements picked up. The system successfully detected approximately 80% of the picked and placed elements. This illustrates the efficacy of picking up and placing text items for the performed tasks. However, in most cases the system did not perfectly capture the intended selection. This was mainly due to the fact that when picking up a selection, some participants placed their start finger one line below the intended line and their end finger in the correct place. Hence, even though close, the intended content was not fully contained in the picked up element.

Clipboard Task

Overall participants thought that the clipboard feature was a useful tool and that the simple interaction style was key. For example, one participant commented: *“I like the ability to pick things out and put them aside for later, it is very useful and being able to do this without leaving the screen that the writing is on is really good.”*

Scenarios that were mentioned for use of the tool included picking out references to look further into; and, grabbing text, to create a simple summary of the document. Several participants thought the ability to capture context about why a note was selected would be a useful feature. Also, several of the participants commented that providing ways of structuring notes captured via different gestures would be an additional useful feature. Finally, linking notes back to the area in document they are from was also suggested.

Haptic Highlighting Feedback Task

Group A participants took considerably longer to locate the search term on a page than those in Group B. To illustrate this, Table 1 shows the mean times taken to locate the first occurrence of three of the search terms under both conditions. An independent-samples t-test was conducted to compare response time for haptic search (Group A) with visual highlighting (Group B). The test showed a highly significant difference with haptic feedback ($M=24.45$, std

dev=24.38) generating slower search times than visual feedback ($M=1.82$, std dev=1.80), $t(412)=13.3$, $p<0.001$.

Search Term (location on page)	Group A Time seconds (std dev)	Group B Time seconds (std dev)
note (top)	6.24 (2.32)	1.78 (0.46)
experience (middle)	19.37 (12.95)	1.49 (0.34)
management (bottom)	18.74 (8.65)	1.34 (0.33)

Table 1: Mean search term location using haptic (Group A) and visual highlighting (Group B).

Clearly with the haptic highlighting, readers have to scan the page in a linear way to actually find the term (the effect of this scanning is illustrated in the shorter search time needed for terms at the top of the page). In Group A there were also instances when participants were flicking through the pages too quickly so that the vibration lagged, resulting in participants looking for words on the wrong page. Participants noted that encoding the number of occurrences and position of a word on a page into the feedback might speed up locating the word. Participants in Group B thought that the visual highlighting was very effective in drawing attention to the area of interest.

Finally, several participants discussed alternative features to on-page feedback that was not linked to the direct entry of search terms to alert the reader of unread relevant references and interesting points in the document.

Search-Plus-Light Task

Fourteen participants commented that the LED feedback was useful for displaying query results without distracting the reader from the document. Thirteen participants commented that they would like both the full query details and results stored for later use, in a similar way to the clipboard, along with the ability to add comments or context to note why the query was made.

Sixteen participants mentioned that lightweight feedback could also be used to present further information. For example, the citation counts of documents retrieved from a search could be used to drive the intensity of the haptic output. Another suggestion was the use of a further haptic pattern to indicate whether the user had already read papers that appeared in the retrieved result list.

CONCLUSIONS AND FUTURE WORK

Hand-based manipulation of digital documents for selecting content for later use (say on a clipboard or search engine) is attractive to users. Additional work might look at adding a repertoire of gestures – e.g. to tag the content indicating its significance. Another interesting direction would be to provide multiple locations around the tablet for objects to be placed (e.g. the top for things for the user, the middle to be shared with others, the bottom for items to be added directly to a bibliography).

In situ lightweight haptic feedback to alert readers about content on the page is less attractive than using direct visual

markup for simple “search within” cases. However, haptic feedback might be valuable in drawing attention to the page without explicitly indicating why it might be of interest. Allowing readers to make inferences about why the system is alerting them to study the page in a more detailed way could lead to serendipitous insights. Similarly, the search-plus-light feedback is something that might be best suited to ambient, automatic feedback based on knowledge of the reader and their research interests. In addition, while we built *NoteDrop* around a tablet computer and a *digital* document, it could be possible to provide some of the facilities for *paper-based* documents using camera input and visual processing to recognize the current page. In this case, where visual mark-up would be harder to implement, the results indicate that a haptic bracelet with embedded LEDs could successfully indicate items of interest.

ACKNOWLEDGMENTS

This work was funded by EPSRC: EP/H042806/1. We thank our colleagues for their input into the *NoteDrop* design and Mike Fraser who helped with statistical analysis.

REFERENCES

1. Dachsel, R. and Sarmad, A.L.S. Interacting with Printed Books Using Digital Pens and Smart Mobile Projection. In *MP2 Workshop at CHI 2011*, (2011).
2. Feldman, A., Tapia, E.M., Sadi, S., Maes, P., and Schmandt, C. ReachMedia: On-the-move interaction with everyday objects. In *Proc. ISWC 2005*, IEEE (2005), 52–59.
3. Gustafson, S., Bierwirth, D., and Baudisch, P. Imaginary interfaces: spatial interaction with empty hands and without visual feedback. In *Proc. UIST 2010*, ACM (2010), 3–12.
4. Hinckley, K., Zhao, S., Sarin, R., Baudisch, P., Cutrell, E., Shilman, M. and Desney, T. InkSeine: In Situ search for active note taking. In *Proc. CHI 2007*, ACM (2007), 251–260.
5. Jones, M., Buchanan, G., Cheng, T.-C., and Jain, P. Changing the pace of search: Supporting “background” information seeking. *JASIST*, John Wiley & Sons, Inc. (2006), 838–842.
6. Marquardt, N., Jota, R., Greenberg, S., Jorge, J. The Continuous Interaction Space: Interaction Techniques Unifying Touch and Gesture On and Above a Digital Surface. In *Proc. Interact 2011*. Springer, 461–476.
7. Marshall, C.C. and Bly, S. Turning the page on navigation. In *Proc. JCDL*, ACM (2005), 225–234.
8. Rekimoto, J. SmartSkin: an infrastructure for freehand manipulation on interactive surfaces. In *Proc. CHI 2002*, ACM (2002), 113–120.
9. Schilit, B.N., Golovchinsky, G., and Price, M.N. Beyond paper: supporting active reading with free form digital ink annotations. In *Proc. CHI 1998*, ACM (1998), 249–256.